RYAN THOMAS McCUNE, EIT

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EDUCATION

Ph.D. Civil Engineering Expected 2027
North Carolina State University Raleigh, NC
Concentration in Coastal Engineering

Master of Civil Engineering 2025
North Carolina State University Raleigh, NC
Concentration in Coastal Engineering

Honors Bachelor of Environmental Engineering with Distinction 2022
University of Delaware Newark, DE
Thesis: Potential Impacts of Soil Aging on TDR Calibrations of Biochar Amended Urban and Coastal Soils

Honors Bachelor of Civil Engineering 2022
University of Delaware Newark, DE
Concentration in Facilities Design and Construction

PROFESSIONAL EXPERIENCE

Graduate Research & Teaching Assistant 2022–present
Department of Civil, Construction, and Environmental Engineering, NC State University Raleigh, NC

- Develop high-resolution coastal flood simulations (SFINCS, ADCIRC) and engineer automated Python workflows on HPC clusters to analyze chronic flood hazards.
- Assemble and deploy custom, low-cost environmental sensor packages (Raspberry Pi, pressure transducers, cameras) and develop rigorous QA/QC data validation frameworks to maintain high-fidelity, publicly accessible flood monitoring datasets.
- Collaborate on the interdisciplinary Sunny Day Flooding Project, integrating physical hydrodynamic modeling with semantic image segmentation and social impact data to characterize regional risk for coastal communities.
- Direct technical training for undergraduate researchers in Python programming and machine learning, and deliver guest lectures and instructional support for a Coastal Engineering course of over 80 students.

Research Intern 2022
United States Geological Survey St. Petersburg, FL

- Developed MATLAB scripts to analyze high-frequency optical data from the USGS coastal camera network, extracting and quantifying wave runup parameters.
- Applied statistical techniques to transform raw imagery into actionable hydrodynamic metrics, supporting regional coastal change research.

Keller Family Senior Writing Fellow 2021–2022
Honors College, University of Delaware Newark, DE

- Directed operational logistics and performance reviews for a university fellowship team as the inaugural Keller Family Senior Fellow.
- Managed the professional development of peer educators, providing strategic mentorship and actionable feedback to improve writing pedagogy and communication coaching.

Engineering Intern 2020–2022
Coastal Resilience Design Studio, University of Delaware Newark, DE

- Collaborated within a multidisciplinary team of landscape architects, policy analysts, and engineers to develop community-informed conceptual designs for coastal public infrastructure.
- Integrated technical engineering constraints with aesthetic and regulatory requirements, culminating in a First Place award in the national Coastal & Estuarine Research Federation (CERF) design competition.

Undergraduate Teaching Assistant 2020–2022
Department of Civil & Environmental Engineering, University of Delaware Newark, DE

- Facilitated supplementary instruction and office hours for students, translating theoretical civil and environmental engineering concepts into practical problem-solving strategies.
- Evaluated technical assignments against established rubrics and provided targeted feedback to bridge gaps in student comprehension.

Undergraduate Research Assistant 2018–2022
Department of Civil & Environmental Engineering, University of Delaware Newark, DE

- Quantified the hydraulic performance of biochar-amended soils for stormwater filtration by calibrating and deploying Time Domain Reflectometry (TDR) sensors.
- Conducted independent laboratory analyses to measure the Electron Reduction Capacity (ERC) of biochar substrates for environmental remediation.
- Executed field data collection for transportation planning studies, deploying JAMAR counters to capture and analyze vehicular traffic volumes across several sites.

Engineering Intern 2020–2021
C.S. Davidson, Inc. York, PA

- Analyzed historical state contract datasets to construct updated unit price schedules, improving the accuracy of municipal project cost estimations and bidding.
- Conducted technical reviews of land development plans to ensure strict compliance with municipal ordinances, zoning codes, and stormwater management regulations.
- Executed precision field surveys and stormwater Best Management Practice (BMP) inspections to verify construction quality and document existing infrastructure.

Writing Fellow 2020–2021
Honors College, University of Delaware Newark, DE

- Completed comprehensive training in writing pedagogy and communication strategy to provide advanced editorial guidance across multiple academic disciplines.
- Partnered with university faculty to mentor a cohort of 20 students per semester, reviewing dozens of academic essays to enhance structural argumentation and clarity.

Munson Fellow 2019–2020
Honors College, University of Delaware Newark, DE

- Advised a residential cohort of over 30 first-year Honors students on degree planning, course selection, and navigating university administration to support institutional retention goals.
- Designed, managed logistics for, and executed community-building events and social programming to foster a highly engaged residential learning environment.

Engineering Intern 2019
Manchester Township York, PA

- Digitized legacy engineering archives to establish a centralized inventory of over 300 stormwater Best Management Practices (BMPs) using the CSDatum platform.
- Optimized municipal maintenance operations by structuring infrastructure data to enable efficient tracking of BMP performance, regulatory compliance, and necessary repairs.

HONORS & AWARDS

National Defense Science & Engineering Graduate Fellowship	2024–2027
ICCE Student Travel Scholarship	2026
KIETS Climate Leaders Program Scholar	2024
AGU Outstanding Student Presenter Award	2023
Honorable Mention, NSF Graduate Research Fellowship Program	2023
2nd Place, NC State EWC Student Poster Competition	2023
Provost Doctoral Fellowship, North Carolina State University	2022
Award of Excellence in Student Collaboration, American Society of Landscape Architects (ASLA)	2022
PA-DE Chapter of the American Society of Landscape Architects Student Honor Award	2022
RJN Foundation Department of Civil & Environmental Engineering Award	2022
1st Place, 2021 Coastal & Estuarine Research Federation Design Competition	2021
Junior Award, Delaware Section of the American Society of Civil Engineers (ASCE)	2021
Honors Enrichment Award, University of Delaware Honors College	2021
Chair's Fellowship, University of Delaware Department of Civil & Environmental Engineering	2021
General Honors Award, University of Delaware Honors College	2020

PRESENTATIONS

Oral Presentations

1. Collins, J., Hino, M., **McCune, R.**, Anarde, K., (2026). *The Price of Paradise: How Communities Tolerate Chronic Coastal Flooding in Rural North Carolina*. Population Association of America Annual Conference. St. Louis, MO, 2026.
2. **McCune, R.**, Anarde, K., Sebastian, A., Collins, J. P., Grimley, L., Hamidi, E., Hino, M., Dietrich, J. C., (Dec. 2025). *The End of the Road: Present and Future Chronic Flood Risk Along Coastal Roadways and Impacts to Community Livability*. Presented at the AGU Fall Meeting. New Orleans, December 2025.

3. **McCune, R.**, Anarde, K., (Aug. 2025). *Evaluation of Chronic Coastal Flooding Inundation of Low-Lying Roadways and Impacts to Community Livability*. Presented at the 3rd Regional (East Coast) Peer Exchange for Sustainable Eco-Resilient Bridges and Structures. Raleigh, August 2025.
4. Collins, J., Hino, M., **McCune, R.**, Anarde, K., Frankenberg, E., (2025). *Tolerating the tide: accommodation and tolerance of chronic coastal flooding in rural North Carolina*. Population Association of America Annual Conference. Washington, D.C., 2025.
5. **McCune, R.**, Anarde, K., Goldstein, E. B., Srebnik, E. R., Thelen, T., Hino, M., (Sept. 2024). *Quantification of chronic coastal flooding using machine learning*. Presented at the International Conference of Coastal Engineering. Rome, September 2024.
6. Hino, M., Anarde, K., **McCune, R.**, Thelen, T., Farquhar, E., Fridell, T., Whipple, T., Woodard, P., (2024). *Incidence and Impacts of Chronic Coastal Flooding in North Carolina*. Invited presentation at the AGU Fall Meeting. Washington, D.C., 2024.
7. **McCune, R.**, Anarde, K., Goldstein, E. B., (Dec. 2023). *Semantic Image Segmentation of Coastal Roadway Inundation*. Presented at the AGU Fall Meeting. San Francisco, December 2023.
8. Muldrow, L., **McCune, R.**, Bruck, J., (Apr. 2022). *Resilient Self-Generative Infrastructure: A Blue Carbon Solution for Coastal Protection in Hampton, VA*. PA-DE ASLA Conference on Landscape Architecture. Wilmington, April 9, 2022.

Poster Presentations

1. **McCune, R.**, Anarde, K., Grimley, L., Collins, J., Hamidi, E., Sebastian, A., Hino, M., Dietrich, C., (Mar. 2026). *Assessing Chronic Flood Risks to Coastal Infrastructure and Communities*. Presented at the NC State Environmental, Water, and Coastal Engineering Symposium. March 6, 2026.
2. **McCune, R.**, Anarde, K., Goldstein, E., Baker, C., (Mar. 2025). *Quantification of Chronic Coastal Flooding: A machine learning-driven approach to water level extraction*. Presented at the NC State Environmental, Water, and Coastal Engineering Symposium. March 21, 2025.
3. **McCune, R.**, Collins, J., Anarde, K., Hino, M., (Sept. 2024). *A Summer Down East: Internship and Research Experiences in Carteret County*. Presented at the KIETS Climate Leaders Symposium 2024. September, 2024.
4. **McCune, R.**, Anarde, K., Hino, M., Frankenburg, E., Amspacher, K., (May 2024). *Rising tides, drowning ditches: Analysis and communication of chronic coastal flooding in rural communities*. Presented at the National Adaptation Forum. Baltimore, May 2024.
5. **McCune, R.**, Anarde, K., Goldstein, E., (Mar. 2024). *Witness to the Rising Tide: Semantic Image Segmentation of Chronic Coastal Flooding*. Presented at the NC State Environmental, Water, and Coastal Engineering Symposium. March 8, 2024.
6. **McCune, R.**, Anarde, K., Goldstein, E., (Mar. 2023). *On-device Machine Learning for Identifying the Spatial Extent of Chronic Coastal Flooding*. Presented at the NC State Environmental, Water, and Coastal Engineering Symposium. March 10, 2023.
7. Anarde, K., Goldstein, E., Bolewitz, J., **McCune, R.**, Gold, A., Hino, M., (Dec. 2022). *On-device machine learning for identifying the spatial extent of chronic coastal floods*. Presented at the International Conference of Coastal Engineering. Sydney, December 5, 2022.
8. **McCune, R.**, Fettke Von Koeckritz, H., Bruck, J., Puleo, J. A., (Oct. 2021). *Fenwick Island Dune Encroachment Monitoring Project*. Presented at the Young Coastal Scientists and Engineers Conference – Americas. Myrtle Beach, October 30, 2021.

Other Invited Lectures or Research Presentations (not listed above)

1. **McCune, R.**, Anarde, K., Hino, M., Nelson, N., (July 2026). *Threats to Coastal Infrastructure from Climate Change*. Presented at the Envisioning Transformations Workshop. July 30, 2026.
2. **McCune, R.**, Anarde, K., Collins, J., Sebastian, A., Hino, M., Dietrich, C., (May 2026). *Below the Waterline: Chronic Flood Risk on Rural Roads and Impacts to Community Livability*. Presented at the 39th International Conference on Coastal Engineering. May 21, 2026.
3. **McCune, R.** (Apr. 2026). *Guest Lecture: Natural and Nature-based Features*. Department of Civil, Construction, and Environmental Engineering. Introduction to Coastal Engineering.
4. **McCune, R.** (Mar. 2025). *Guest Lecture: Natural and Nature-based Features*. Department of Civil, Construction, and Environmental Engineering. Introduction to Coastal Engineering.
5. **McCune, R.** (Mar. 2024). *Guest Lecture: Natural and Nature-based Features*. Department of Civil, Construction, and Environmental Engineering. Introduction to Coastal Engineering.
6. **McCune, R.** (Mar. 2023). *Guest Lecture: Natural and Nature-based Features*. Department of Civil, Construction, and Environmental Engineering. Introduction to Coastal Engineering.

PUBLICATIONS

Peer-Reviewed Journal Articles (Published or Accepted)

1. Collins, J. P., **McCune, R.**, Hino, M., Anarde, K., Amspacher, K., (Aug. 2026). “The Price of Paradise: How Communities Tolerate Chronic Coastal Flooding in Rural North Carolina”. In: *Frontiers in Climate* 8. ISSN: 2624-9553. DOI: `10.3389/fclim.2026.1880823`.
2. Hino, M., Anarde, K., Fridell, T., **McCune, R.**, Thelen, T., Farquhar, E., Woodard, P., Whipple, A., (June 2025). “Land-based sensors reveal high frequency of coastal flooding”. en. In: *Communications Earth & Environment* 6.1, pp. 1–6. ISSN: 2662-4435. DOI: `10.1038/s43247-025-02326-w`.
3. Naquin, K., Adams, D. R., Bailey, M. M., Brown, L., Diez, M., Kanipe, J., **McCune, R.**, Thelen, T., Hunter, D. L., Cooper, C. B., (Apr. 2025). “Not Empty Rain Gauges: Experienced Hobbyists Fulfilled in a Contributory Project”. In: *Citizen Science: Theory and Practice*. DOI: `10.5334/cstp.774`.

Peer-Reviewed Journal Articles (Under Review & In Preparation)

1. **McCune, R.**, Anarde, K., Goldstein, E., Baker, C., (in preparation). “Quantification of chronic coastal flooding: a machine-learning driven approach to water level extraction”. In preparation for Water Resources Research.

Datasets

1. Goldstein, E. B., Buscombe, D., Budavi, P., Favela, J., Fitzpatrick, S., Gabbula, S. R. A. K., Ku, V., Lazarus, E. D., **McCune, R.**, Shah, M., Sigdel, R., Tagner, S., (Nov. 2022). *Segmentation Labels for Emergency Response Imagery from Hurricane Barry, Delta, Dorian, Florence, Isaias, Laura, Michael, Sally, Zeta, and Tropical Storm Gordon*. Version v1. Zenodo. DOI: `10.5281/zenodo.7268083`.

PROFESSIONAL DEVELOPMENT

Short Course: Hydraulic Structures and Storm Surge Barriers, Galveston, TX	2026
COPRI Leadership Summit, Reston, VA	2026
Community Surface Dynamics Modeling System Annual Meeting, Boulder, CO	2025
Earth Surface Processes Institute, Boulder, CO	2025

Short Course: Modeling Compound Flooding with SFINCS, Rome, IT	2024
From Ice Sheets to the Coast: Sea-Level Rise Impacts Workshop, Houston, TX	2024
5th NOAA AI Workshop, virtual	2023
Coastal Imaging Research Network Workshop, Duck, NC	2023
Blue Economy Workshop, Morehead City, NC	2023

PROFESSIONAL SERVICE

Chair | EWC Seminar Visiting Student Logistics Committee, NC State University	2026
Student Ambassador | NC State Climate and Sustainability Academy	2025
Session Chair and Organizer | "The MacGyver Session" Ocean Sciences Poster Session, AGU Annual Meeting	2025
Member | EWC Seminar Visiting Student Logistics Committee, NC State University	2025
Organizer and Moderator | Panel: Community Responses to Chronic Flooding and Sea-Level Rise Impacts, North Carolina Coastal Conference	2024
Chair | EWC Seminar Food Committee, NC State University	2024
Member | EWC Seminar Food Committee, NC State University	2023
Student Member | Provost Search Committee, University of Delaware	2022
Engineering Ambassador | College of Engineering, University of Delaware	2022
Engineering Ambassador | Dept. of Civil & Environmental Engineering, University of Delaware	2021–2022
Honors College Ambassador | University of Delaware	2019–2022

Professional Affiliations

Member, American Society of Civil Engineers (ASCE)
Member, American Shore and Beach Preservation Association (ASBPA)
Member, American Geophysical Union (AGU)
Member, Tau Beta Pi Engineering Honor Society
Member, Chi Epsilon Civil Engineering Honor Society

Manuscript Reviewer

Geoscientific Model Development (1)

LEADERSHIP

Vice President | Coasts, Oceans, Ports & Rivers Institute (COPRI) Student Chapter	2026–present
President | Coasts, Oceans, Ports & Rivers Institute (COPRI) Student Chapter	2025–2026
Vice President | Coasts, Oceans, Ports & Rivers Institute (COPRI) Student Chapter	2023–2024
Vice President | Environmental Engineering Student Association	2020–2022
Parliamentarian | Phi Sigma Pi National Honor Fraternity Alpha Eta Chapter	2021